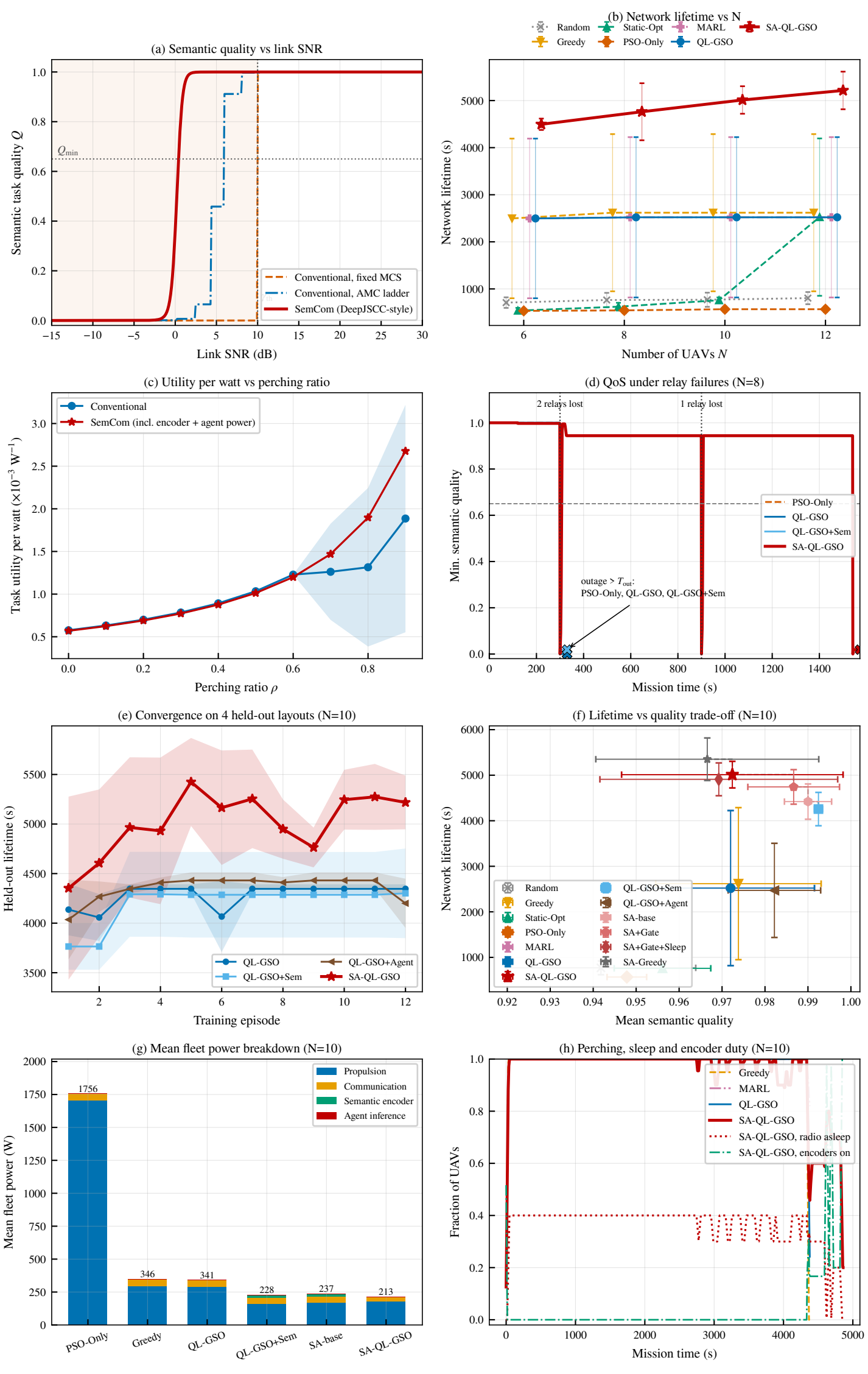
(a) Semantic quality vs link SNR
Semantic task quality Q
Link SNR (dB)
$Q_{\min}$
Conventional, fixed MCS
Conventional, AMC ladder
SemCom (DeepJSCC-style)
(b) Network lifetime vs N
Random
Greedy
Static-Opt
PSO-Only
MARL
QL-GSO
SA-QL-GSO
Network lifetime (s)
Number of UAVs N
(c) Utility per watt vs perching ratio
Conventional
SemCom (incl. encoder + agent power)
Task utility per watt ($\times 10^{-3}$ W^{-1})
Perching ratio ρ
(d) QoS under relay failures (N=8)
2 relays lost
1 relay lost
outage > T_{out}:
PSO-Only, QL-GSO, QL-GSO+Sem
PSO-Only
QL-GSO
QL-GSO+Sem
SA-QL-GSO
Min. semantic quality
Mission time (s)
(e) Convergence on 4 held-out layouts (N=10)
QL-GSO
QL-GSO+Sem
QL-GSO+Agent
SA-QL-GSO
Held-out lifetime (s)
Training episode
(f) Lifetime vs quality trade-off (N=10)
Random
Greedy
Static-Opt
PSO-Only
MARL
QL-GSO
SA-QL-GSO
QL-GSO+Sem
QL-GSO+Agent
SA-base
SA+Gate
SA+Gate+Sleep
SA-Greedy
Network lifetime (s)
Mean semantic quality
(g) Mean fleet power breakdown (N=10)
Propulsion
Communication
Semantic encoder
Agent inference
Mean fleet power (W)
1756
346
341
228
237
213
PSO-Only
Greedy
QL-GSO
QL-GSO+Sem
SA-base
SA-QL-GSO
(h) Perching, sleep and encoder duty (N=10)
Greedy
MARL
QL-GSO
SA-QL-GSO
SA-QL-GSO, radio asleep
SA-QL-GSO, encoders on
Fraction of UAVs
Mission time (s)